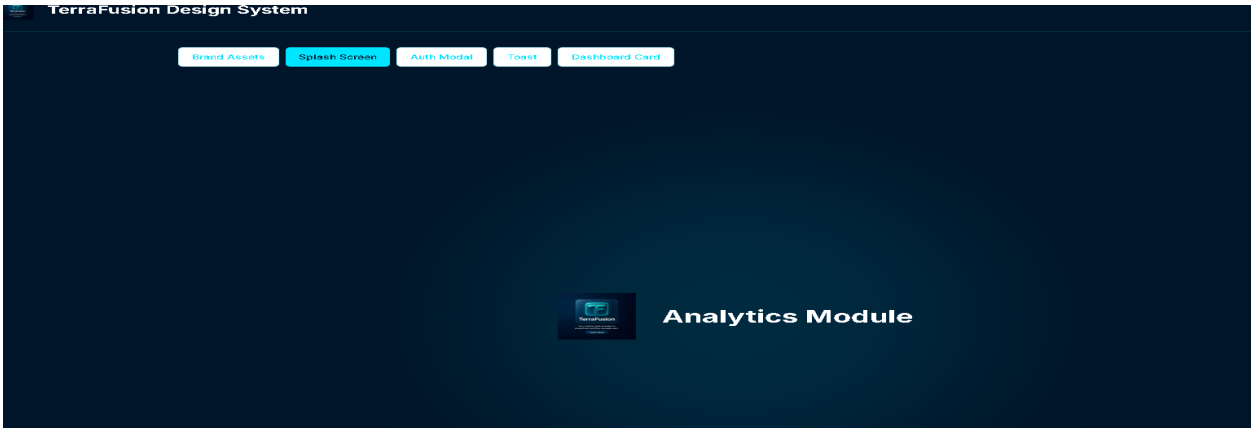


Benton County — Vertical Equity Indicators

2010–2025 (from available sales extracts) • PRD / PRB / Ratio-by-Tier

Purpose

This section quantifies vertical equity (price-related bias) in assessment ratios over time. It complements level (Median Ratio) and uniformity (COD) by testing whether higher- and lower-value properties are treated proportionately through PRD, PRB, and ratio-by-sale-price tier diagnostics.

Key Indicators

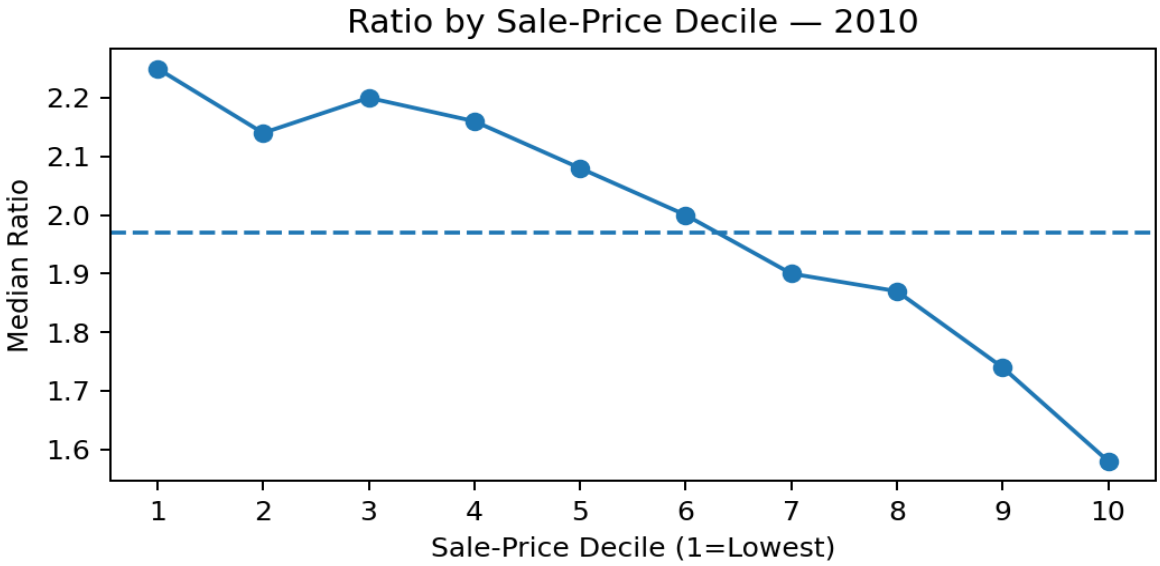
PRD: Price-Related Differential (mean ratio ÷ weighted mean ratio). Values near 1.00 indicate limited price-related bias; persistent deviation can indicate regressivity/progressivity.
PRB: Price-Related Bias slope from ln(ratio) ~ ln(price). Negative slope suggests ratios decrease as value increases (regressive assessment risk).
Tier Plot: Median ratios across sale-price deciles show where bias concentrates.

Annual Vertical Equity Summary (Benton County)

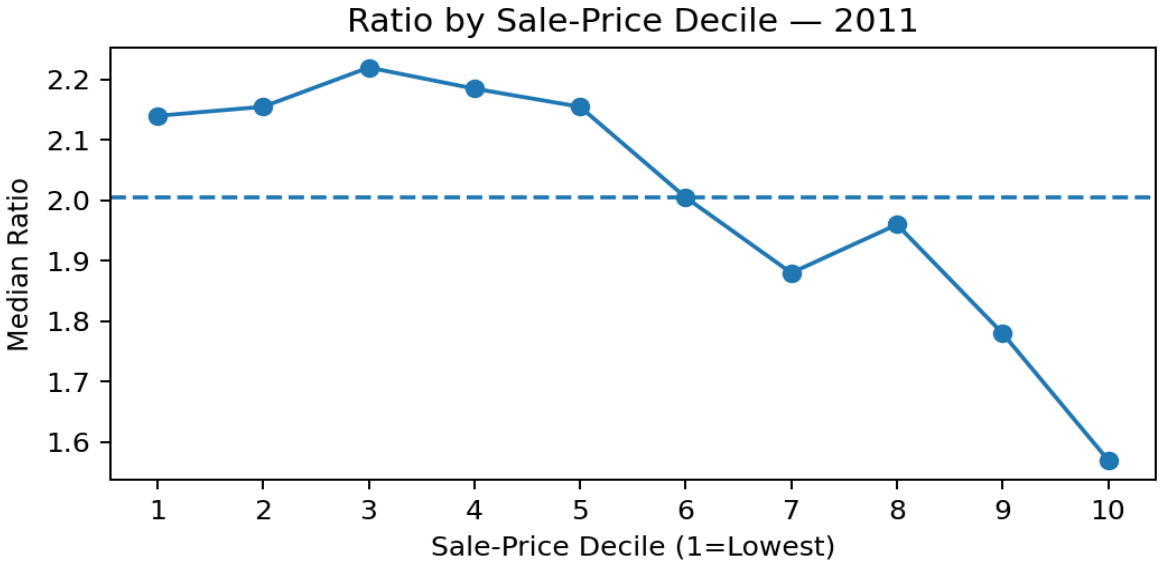
Year	N	Median Ratio	COD	PRD	PRB slope
2010	1030	1.970	16.51	1.092	-0.330
2011	936	2.005	14.85	1.227	-0.239
2012	1149	2.000	17.35	1.277	-0.326
2013	1444	1.970	14.70	1.184	-0.248
2014	1526	1.925	17.00	1.142	-0.275
2015	2002	1.850	15.46	1.147	-0.240
2016	2201	1.720	13.57	1.066	-0.175
2017	2338	1.575	13.98	1.061	-0.158
2018	2449	1.450	12.33	1.032	-0.120
2019	2351	1.360	10.79	1.029	-0.098
2020	2869	1.260	11.27	1.029	-0.116
2021	2532	1.080	13.11	1.049	-0.114
2022	2351	0.950	10.75	1.010	-0.036
2023	2168	0.950	7.82	1.000	0.015
2024	2768	0.940	6.87	1.008	-0.013
2025	3070	0.910	12.31	1.049	-0.057

Appendix: Ratio by Sale-Price Decile (Median Ratio per Decile)

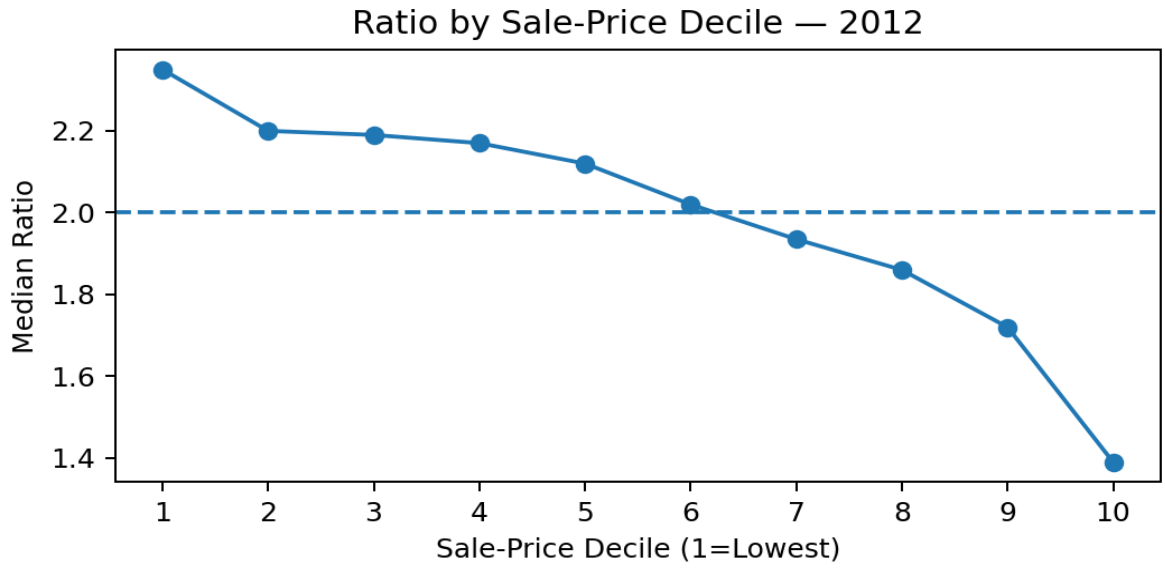
2010



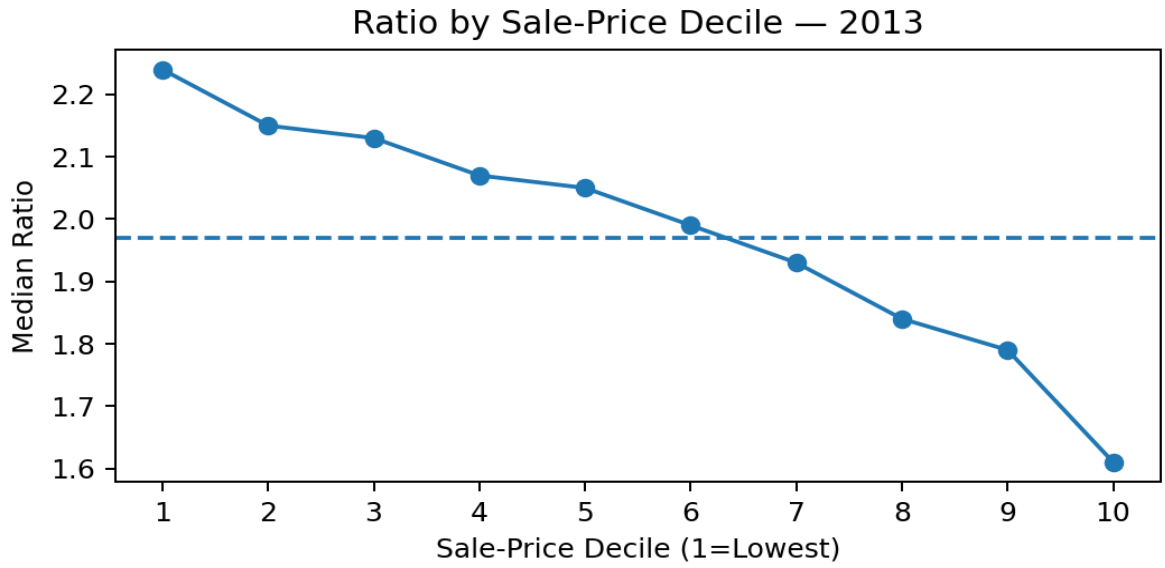
2011



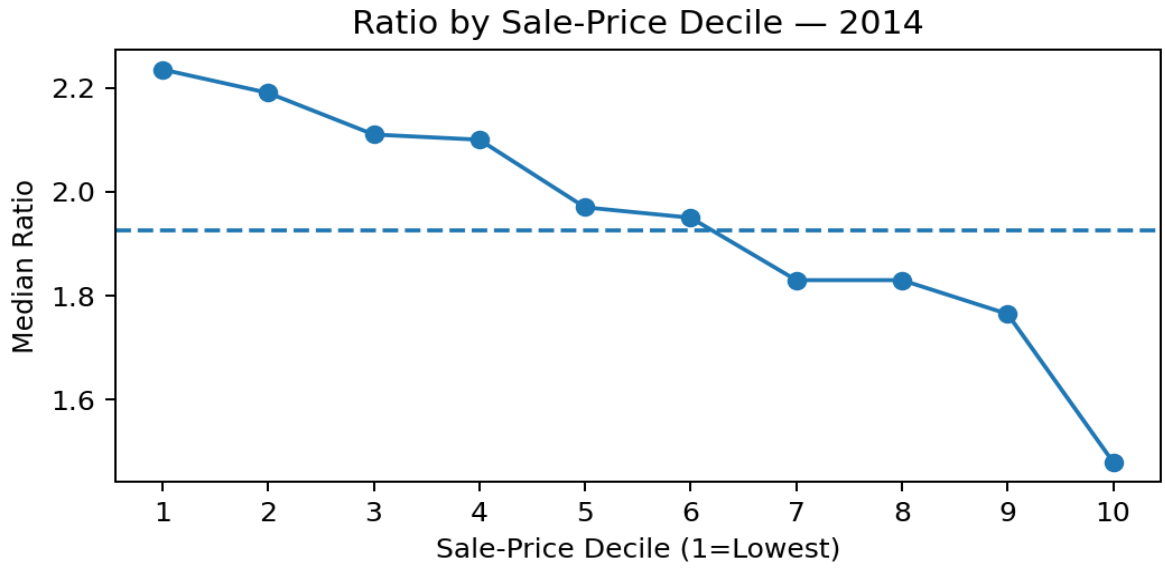
2012


Ratio by Sale-Price Decile — 2012
Median Ratio
2.2
2.0
1.8
1.6
1.4
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

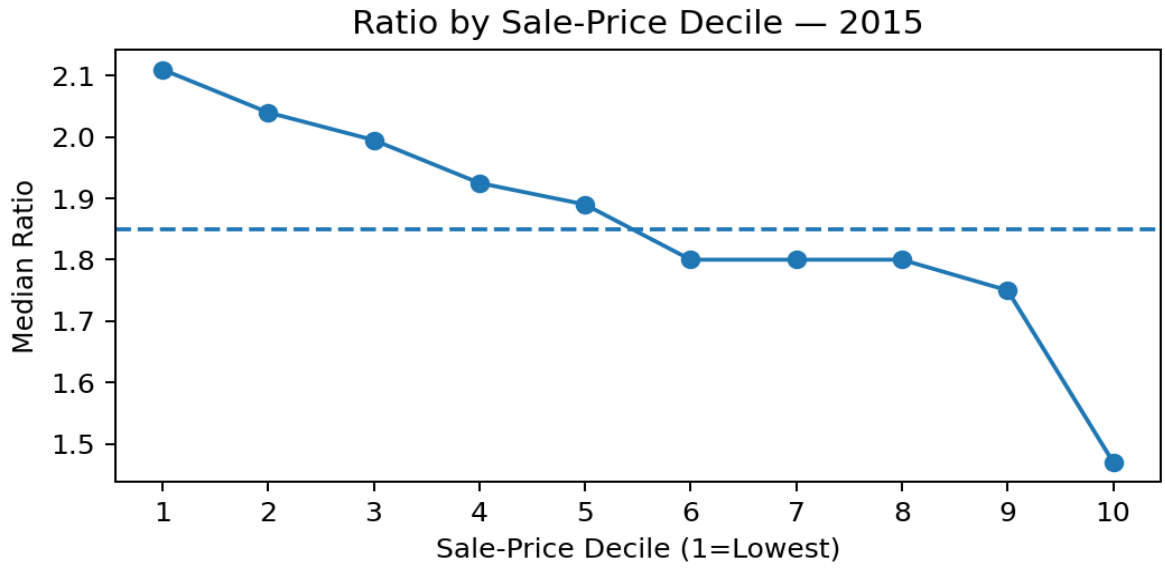
2013


Ratio by Sale-Price Decile — 2013
Median Ratio
2.2
2.1
2.0
1.9
1.8
1.7
1.6
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

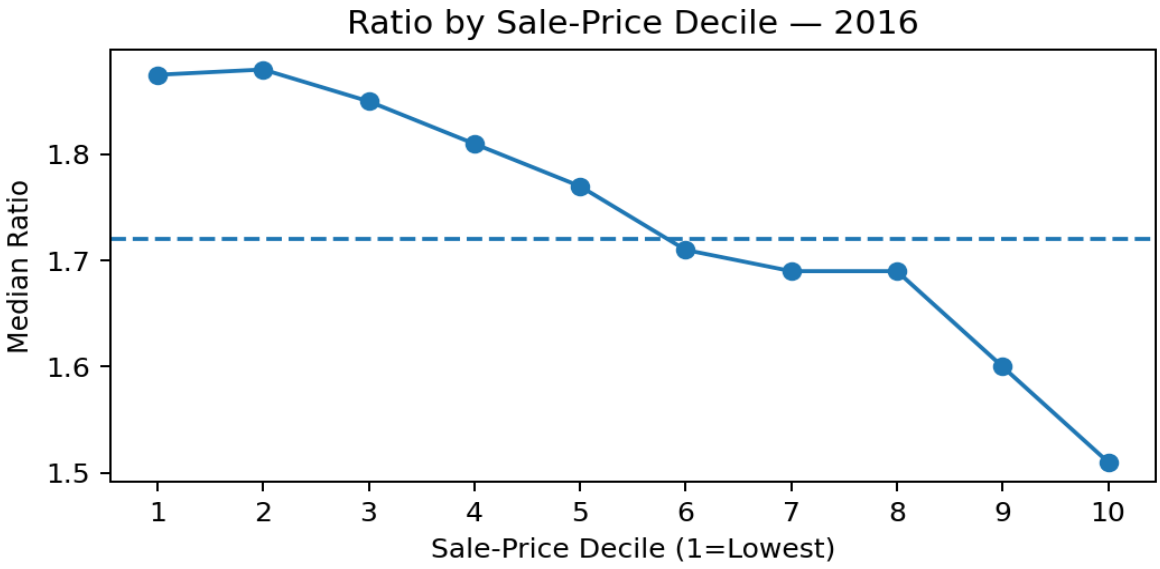
2014


Ratio by Sale-Price Decile — 2014
Median Ratio
2.2
2.0
1.8
1.6
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

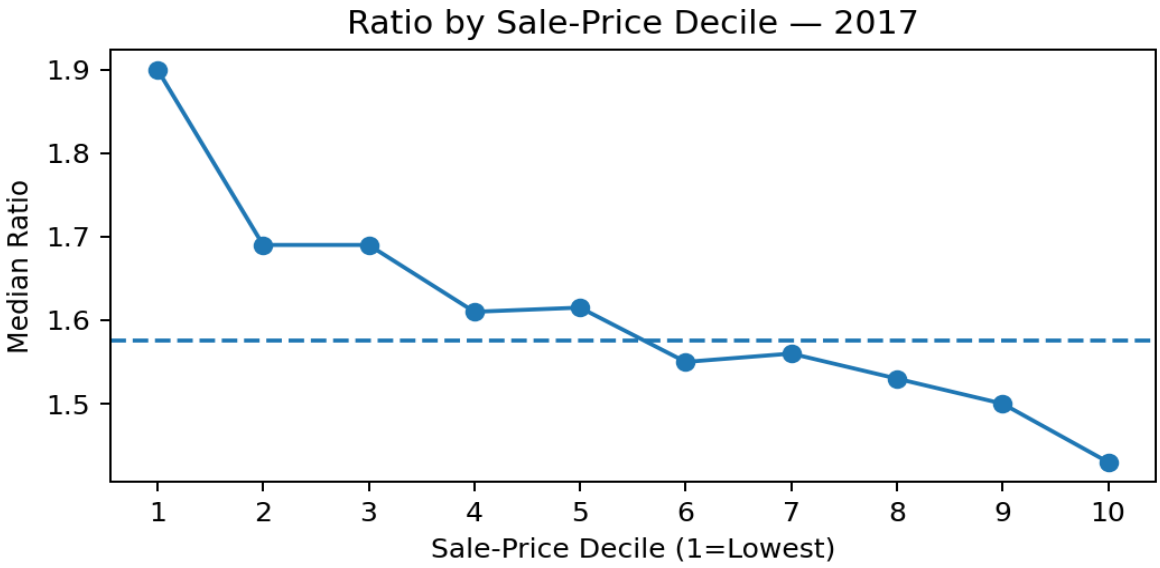
2015


Ratio by Sale-Price Decile — 2015
Median Ratio
2.1
2.0
1.9
1.8
1.7
1.6
1.5
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

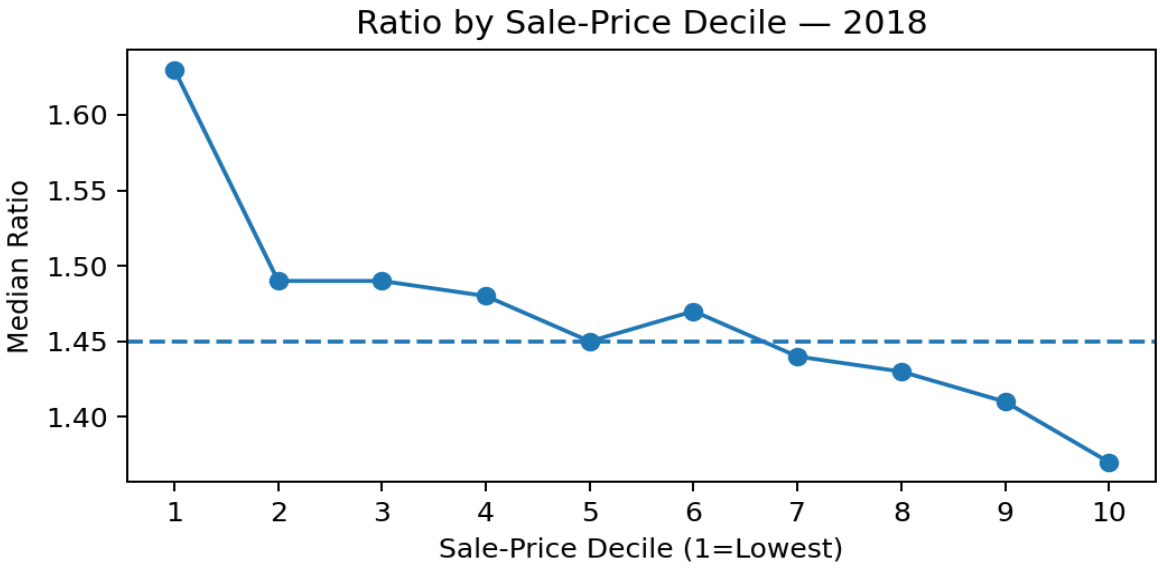
2016


Ratio by Sale-Price Decile — 2016
Median Ratio
1.8
1.7
1.6
1.5
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

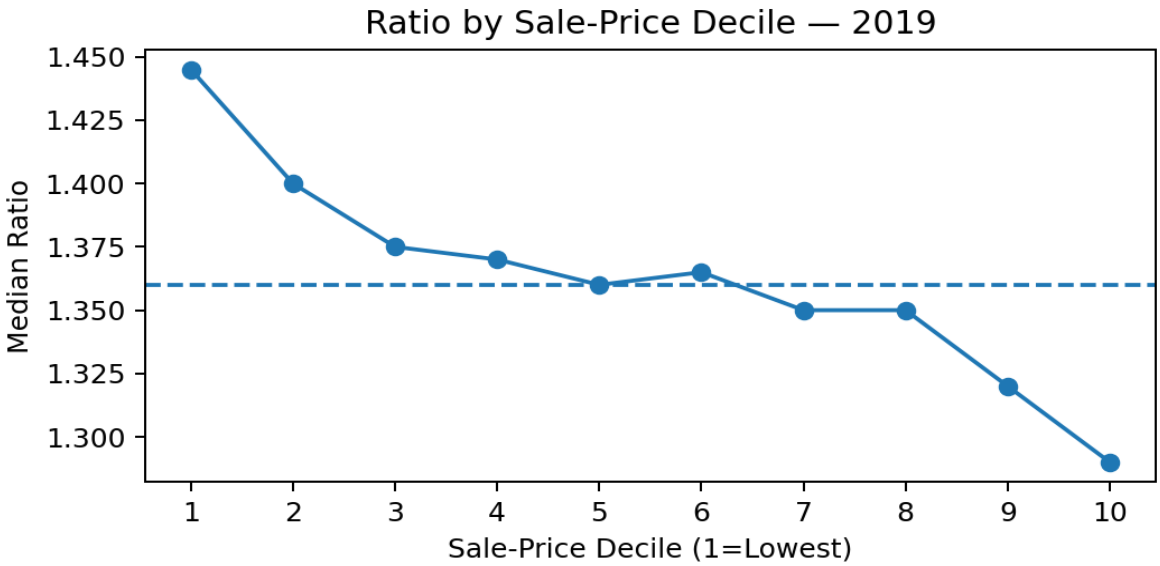
2017


Ratio by Sale-Price Decile — 2017
Median Ratio
1.9
1.8
1.7
1.6
1.5
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

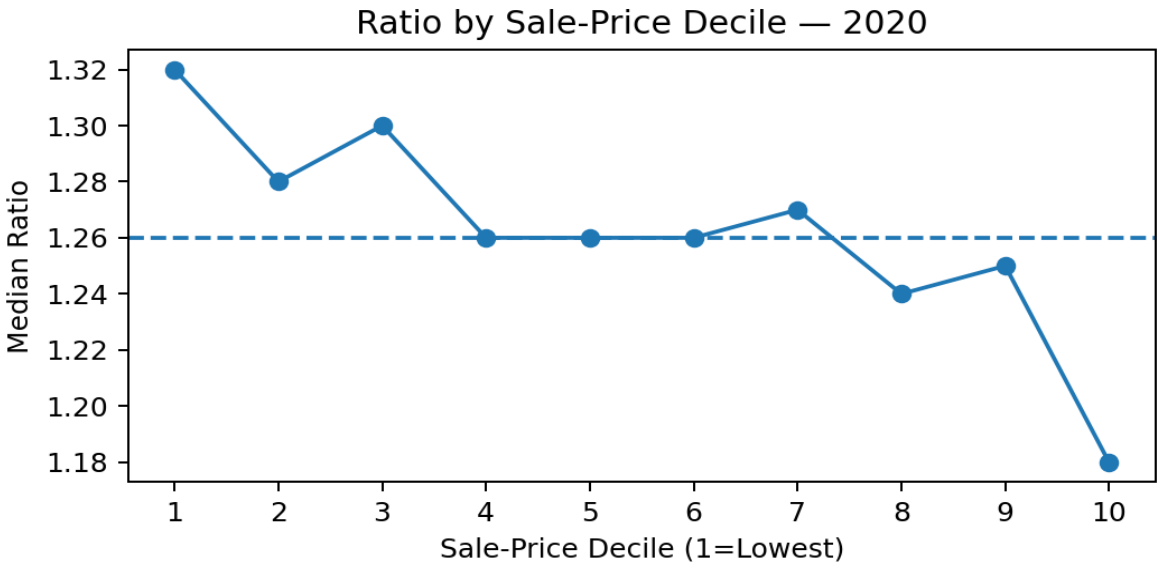
2018


Ratio by Sale-Price Decile — 2018
Median Ratio
1.60
1.55
1.50
1.45
1.40
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

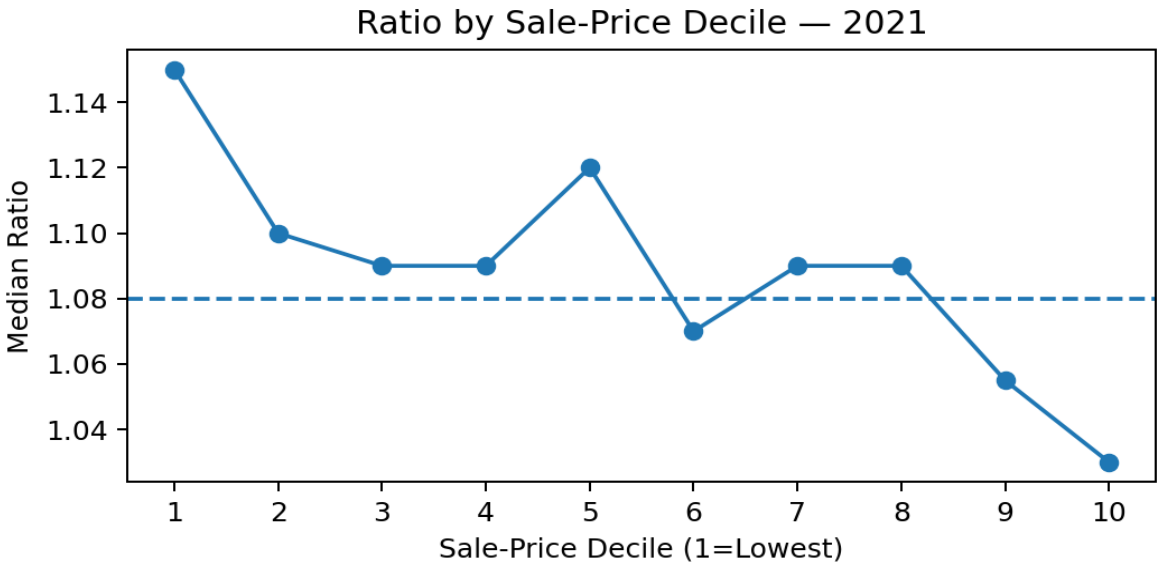
2019


Ratio by Sale-Price Decile — 2019
Median Ratio
1.450
1.425
1.400
1.375
1.350
1.325
1.300
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

2020


Ratio by Sale-Price Decile — 2020
Median Ratio
1.32
1.30
1.28
1.26
1.24
1.22
1.20
1.18
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

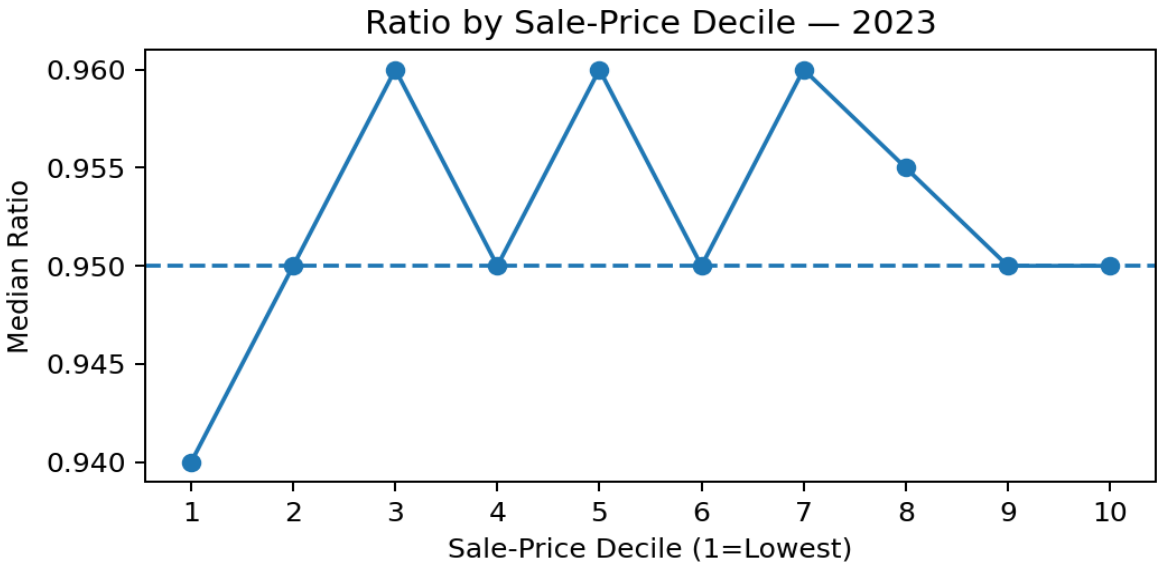
2021


Ratio by Sale-Price Decile — 2021
Median Ratio
1.14
1.12
1.10
1.08
1.06
1.04
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

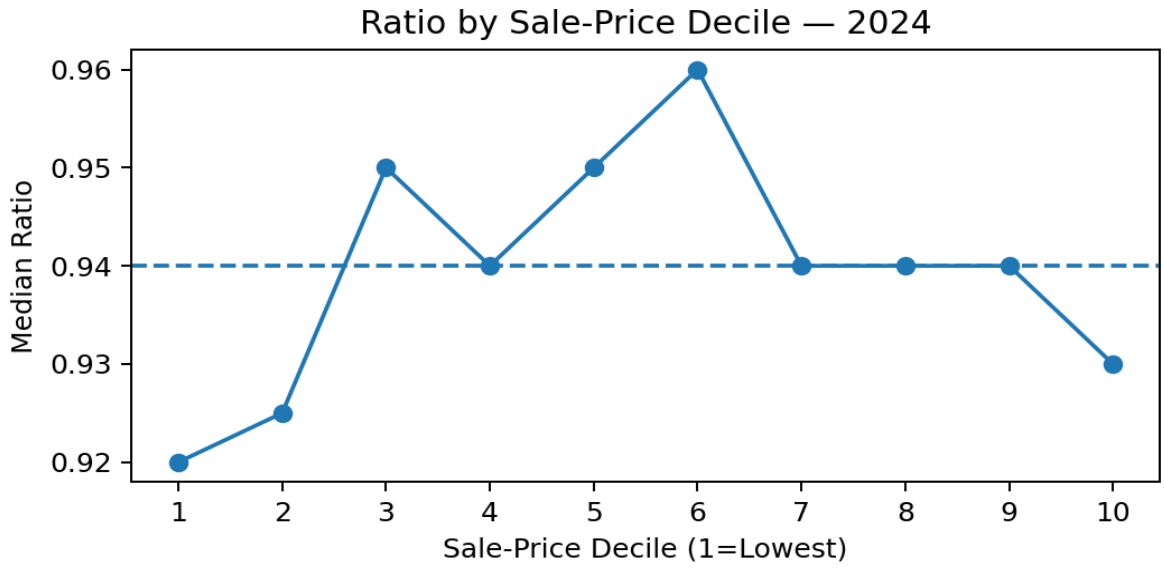
2022

Ratio by Sale-Price Decile — 2022
Median Ratio
0.970
0.965
0.960
0.955
0.950
0.945
0.940
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

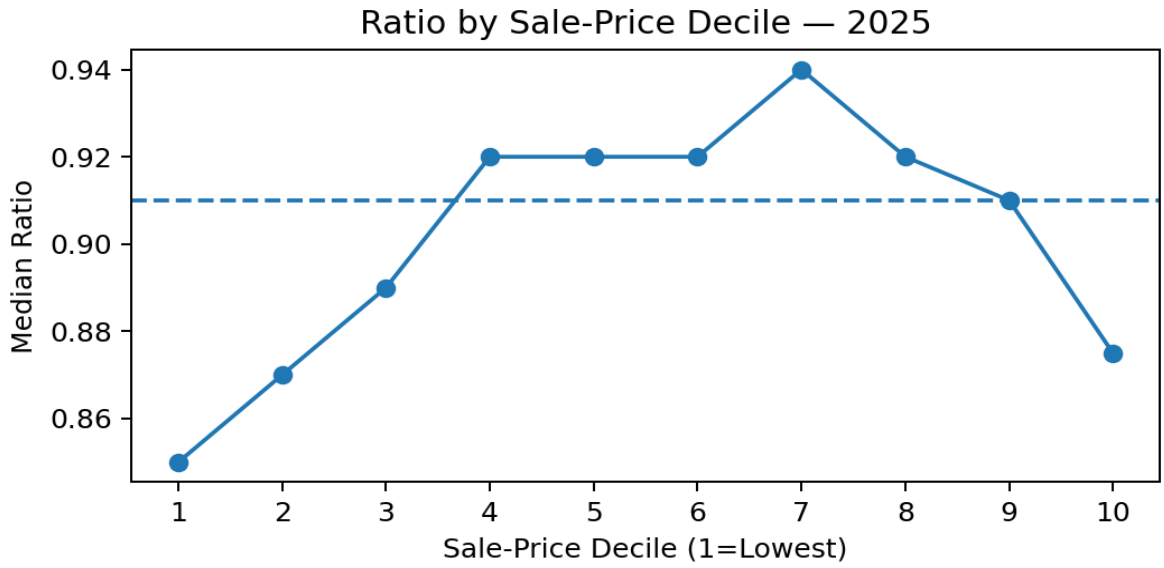
2023


Ratio by Sale-Price Decile — 2023
Median Ratio
0.960
0.955
0.950
0.945
0.940
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

2024


Ratio by Sale-Price Decile — 2024
Median Ratio
0.96
0.95
0.94
0.93
0.92
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)

2025


Ratio by Sale-Price Decile — 2025
Median Ratio
0.94
0.92
0.90
0.88
0.86
1
2
3
4
5
6
7
8
9
10
Sale-Price Decile (1=Lowest)